

KLH UNIVERSITY

Department of Computer Science and Engineering

A PROJECT REPORT ON

Portfolio Risk-Factor Analysis Using PCA

Dimensionality Reduction for Portfolio Risk Management

Submitted by

B Shammruth **Roll No: 2520030191**

U. Praneeth **Roll No: 2520030188**

Under the guidance of

Shaik Asif Sir

Department of CSE, KLH University

Academic Year: 2026-2027

Contents

Abstract

1. Problem Statement
2. Case-Based Scenario
3. Objectives
4. Stock-Market Dataset
5. Exploratory Data Analysis and Preprocessing
6. Correlation Analysis
7. PCA Algorithm Theory
8. PCA Mathematical Formulas
9. Step-by-Step PCA Implementation in Python
10. Explained Variance and Cumulative Variance
11. Component and Loadings Analysis
12. Graphs and Results
13. Interpretation of Risk Factors
14. Conclusion and Future Scope
15. References

Appendix: Viva Questions and Answers

Note: Boxes marked "Insert Colab output" and blanks marked ____ are to be filled with the actual results from the Google Colab notebook. No results have been assumed in this report.

Abstract

A portfolio usually holds many stocks whose returns move together. Studying each stock alone hides the few common forces that decide most of the portfolio risk. This project applies Principal Component Analysis (PCA) to the daily returns of a stock-market dataset to reduce many correlated variables into a small number of uncorrelated risk factors.

The work covers data exploration, daily return calculation, missing value and outlier handling, standardization, correlation analysis, PCA using both manual eigen-decomposition and scikit-learn, study of explained and cumulative variance, and interpretation of component loadings as risk factors. The implementation is done in Python on Google Colab.

1. Problem Statement

Investors and risk managers hold portfolios with tens or hundreds of stocks. Stock returns are strongly related because they respond to the same market, sector and economic news. As a result, the returns matrix has many variables but far fewer independent sources of movement.

Problem: How can a large set of correlated stock returns be reduced to a small set of uncorrelated risk factors, while keeping most of the information about portfolio risk?

Working with all stocks directly causes three difficulties: (i) the covariance matrix is large and noisy, (ii) repeated information hides the true risk drivers, and (iii) the results are hard to explain to a non-technical decision maker. PCA is a standard dimensionality reduction method that addresses all three.

2. Case-Based Scenario

Scenario. A risk analyst works at an asset-management firm. The firm holds a portfolio of many listed stocks and has their daily prices. The portfolio manager asks: "Which few underlying factors explain most of our daily risk, and which stocks drive each factor?"

Stage	Description
Situation	A portfolio of many stocks, with daily price data available.
Difficulty	Stocks in the same sector or in the same market move together, so tracking every stock separately gives noisy and repeated information.
Manager's question	Which few factors explain most of the daily risk?
Analyst's task	Compute daily returns, clean and standardize them, and apply PCA to find the main risk factors.
Expected outcome	A small number of principal components that explain most of the variance, with each component linked to a clear risk meaning (for example market-wide risk or sector risk).

Table 1: Case scenario summary

3. Objectives

1. To explore the stock-market dataset using exploratory data analysis (EDA).
2. To calculate daily returns from closing prices.
3. To handle missing values and outliers.
4. To standardize the returns and study correlation between stocks.
5. To apply PCA and study explained and cumulative variance.
6. To interpret the principal components as portfolio risk factors.

The methodology follows the pipeline shown below.



Figure 1: Methodology pipeline

4. Stock-Market Dataset

The project uses a daily stock-market dataset with one row per stock per trading day. The fields are shown below.

Field	Meaning	Use in this project
timestamp	Trading date	Time index
symbol	Stock ticker	Becomes a column after pivoting
open, high, low	Price levels during the day	EDA and daily range
close	Closing price	Daily return calculation
previous close	Closing price of previous day	Cross-check of returns
volume	Number of shares traded	Liquidity in EDA
turnover	Value of shares traded	Liquidity in EDA

Table 2: Dataset fields

Dataset facts (fill from Colab): rows $\times$ columns = ____; number of stocks = ____; date range = ____ to ____; data source = ____.

5. Exploratory Data Analysis and Preprocessing

5.1 EDA program

The first step is to load the data and study its structure, summary statistics and missing values.

```
import pandas as pd, numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('stocks.csv', parse_dates=['timestamp'])

print(df.shape)          # rows and columns
```

```
df.info()          # data types
print(df.describe()) # summary statistics
print(df.isnull().sum()) # missing values per column
print(df['symbol'].nunique(), 'stocks')
```

Figure 2: Dataset structure and summary statistics

5.2 Daily return calculation

Prices are non-stationary and are on different scales for different stocks. Returns are comparable across stocks, so PCA is applied on returns, not prices. The daily return is the percentage change in the closing price:

$$r(t) = (P(t) - P(t-1)) / P(t-1)$$

```
# rows = dates, columns = stocks
prices = df.pivot(index='timestamp', columns='symbol', values='close')
returns = prices.pct_change().dropna(how='all')

prices.plot(figsize=(10,4), legend=False, title='Closing prices')
plt.show()
returns.stack().hist(bins=60)
plt.title('Distribution of daily returns'); plt.show()
```

Figure 3: Closing price trend

Figure 4: Distribution of daily returns

5.3 Missing value handling

Stocks with too many missing days are removed, because filling large gaps would add false information. Small gaps are filled with the previous day's value (forward fill), and any remaining empty rows are dropped.

```
# keep stocks with at least 90% data available
returns = returns.dropna(thresh=int(0.9*len(returns)), axis=1)
returns = returns.ffill().dropna()
print(returns.isnull().sum().sum(), 'missing values left')
```

5.4 Outlier handling

Extreme returns can be real events (for example a market crash) or data errors. They can pull the principal components towards themselves. The IQR rule is used to detect outliers and extreme values are clipped (winsorized) instead of being deleted, so no dates are lost.

$$\text{Lower} = Q1 - k \cdot \text{IQR}, \text{Upper} = Q3 + k \cdot \text{IQR}, \text{IQR} = Q3 - Q1$$

```
Q1, Q3 = returns.quantile(.25), returns.quantile(.75)
IQR = Q3 - Q1
returns = returns.clip(Q1 - 3*IQR, Q3 + 3*IQR, axis=1) # k = 3
```

Number of values clipped (fill from Colab): ____.

5.5 Standardization

PCA looks for directions of maximum variance. If a stock has a larger variance than others, it would dominate the first component only because of scale. Standardization gives every stock mean 0 and variance 1.

$$z = (x - \mu) / \sigma$$

```
from sklearn.preprocessing import StandardScaler
X = StandardScaler().fit_transform(returns)
print(X.mean(axis=0).round(6)[:5], X.std(axis=0)[:5])
```

6. Correlation Analysis

Correlation shows how strongly two stocks move together. PCA works well only when variables are correlated, so this step checks whether PCA is suitable. On standardized data, the covariance matrix and the correlation matrix are the same.

$$\rho(A,B) = \text{Cov}(A,B) / (\sigma_A \cdot \sigma_B)$$

```
corr = returns.corr()
plt.figure(figsize=(9,7))
sns.heatmap(corr, cmap='coolwarm', center=0)
plt.title('Correlation of daily returns'); plt.show()

# average pairwise correlation
mask = np.triu(np.ones(corr.shape, dtype=bool), k=1)
print('Average correlation:', corr.values[mask].mean())
```

Figure 5: Correlation heatmap of daily returns

Observation (write from your heatmap): _____.

7. PCA Algorithm Theory

Principal Component Analysis converts a set of correlated variables into a new set of uncorrelated variables called principal components. Each component is a weighted combination of the original stocks.

- The first component (PC1) is the direction of largest variance in the data.
- The second component (PC2) has the largest variance among directions perpendicular to PC1, and so on.
- All components are orthogonal, so they are uncorrelated.
- Keeping only the first k components reduces the dimension from p stocks to k factors with little loss of information.

Algorithm steps

Step	Action
1	Standardize the returns matrix Z (n days × p stocks).
2	Compute the covariance (correlation) matrix C.
3	Find eigenvalues and eigenvectors of C.
4	Sort eigenvalues from largest to smallest.
5	Compute the explained variance ratio of each component.
6	Choose the number of components k.
7	Project the data: PC = Z · W _k .

Table 3: PCA algorithm steps

Simple two-stock example (for understanding)

Take two standardized stocks with correlation ρ . The correlation matrix is $C = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. Solving $\det(C - \lambda I) = 0$ gives the eigenvalues $\lambda_1 = 1 + \rho$ and $\lambda_2 = 1 - \rho$. The eigenvector for λ_1 is $(1, 1)/\sqrt{2}$, so PC1 is the average of the two stocks. The eigenvector for λ_2 is $(1, -1)/\sqrt{2}$, so PC2 is their difference. PC1 explains $(1 + \rho)/2$ of the total variance. When ρ is close to 1, PC1 alone explains almost everything. This is exactly why highly correlated stocks can be reduced to fewer factors.

8. PCA Mathematical Formulas

Quantity	Formula
Daily return	$r(t) = (P(t) - P(t-1)) / P(t-1)$
Standardization	$z = (x - \mu) / \sigma$
Covariance matrix	$C = Z^T Z / (n - 1)$
Eigen equation	$C v = \lambda v$
Characteristic equation	$\det(C - \lambda I) = 0$
Explained variance ratio	$EVR(i) = \lambda_i / (\lambda_1 + \lambda_2 + \dots + \lambda_p)$
Cumulative variance	$CVR(k) = EVR(1) + EVR(2) + \dots + EVR(k)$
Principal component scores	$PC = Z \cdot W_k$ (W_k = first k eigenvectors)
Portfolio variance	$\sigma_p^2 = w^T \Sigma w$ (w = weights, Σ = covariance)

Table 4: Key formulas

Here λ (eigenvalue) is the variance captured by a component, and v (eigenvector) gives the loadings, that is, the weights of each stock in that component. The sum of all eigenvalues equals the total variance, which is p for standardized data.

9. Step-by-Step PCA Implementation in Python**Method A: Manual eigen-decomposition**

```
# Step 2: covariance (= correlation) matrix of standardized data
C = np.cov(X.T)

# Step 3: eigenvalues and eigenvectors
vals, vecs = np.linalg.eigh(C)

# Step 4: sort in descending order
idx = np.argsort(vals)[::-1]
vals, vecs = vals[idx], vecs[:, idx]

# Step 5: explained variance ratio
evr_manual = vals / vals.sum()
cum_manual = np.cumsum(evr_manual)

# Step 7: projection on first k components (k chosen later)
# scores = X @ vecs[:, :k]
```

Method B: scikit-learn

```
from sklearn.decomposition import PCA

pca = PCA()
pcs = pca.fit_transform(X) # keep all components

evr = pca.explained_variance_ratio_
cum = np.cumsum(evr)

# Step 6: smallest k reaching 90% cumulative variance
k = np.argmax(cum >= 0.90) + 1
print('Components needed for 90% variance:', k)

pcs_df = pd.DataFrame(pcs[:, :k], index=returns.index,
                      columns=[f'PC{i+1}' for i in range(k)])

# verification: both methods should agree
print(np.allclose(evr, evr_manual))
```

Both methods must give the same explained variance. Signs of eigenvectors may differ, which does not change the meaning of a component.

Figure 6: Implementation output

10. Explained Variance and Cumulative Variance

Explained variance shows how much of the total variation is captured by each component. Cumulative variance shows the total captured by the first k components. Three common rules are used to choose k .

- **Scree plot (elbow):** keep components before the curve flattens.
- **Kaiser rule:** keep components with eigenvalue greater than 1 on standardized data.
- **Cumulative threshold:** keep enough components to reach about 90% of variance.

```
fig, ax = plt.subplots(1, 2, figsize=(12,4))
ax[0].plot(range(1, len(evr)+1), evr, 'o-')
ax[0].set_title('Scree plot'); ax[0].set_xlabel('Component')
ax[1].plot(range(1, len(cum)+1), cum, 'o-')
ax[1].axhline(0.90, color='r', ls='--')
ax[1].set_title('Cumulative explained variance')
plt.show()
```

Component	Eigenvalue	Explained variance (%)	Cumulative variance (%)
PC1	_____	_____	_____
PC2	_____	_____	_____
PC3	_____	_____	_____
PC4	_____	_____	_____
PC5	_____	_____	_____

Table 5: Variance table (fill from `pca.explained_variance_`, `evr` and `cum`)

Figure 7: Scree plot

Figure 8: Cumulative explained variance

11. Component and Loadings Analysis

Loadings are the entries of the eigenvectors. A loading tells how strongly a stock contributes to a component. Stocks with large positive or large negative loadings define the meaning of that component.

```
loadings = pd.DataFrame(pca.components_[:3].T,
                        index=returns.columns,
                        columns=['PC1', 'PC2', 'PC3'])

print(loadings.sort_values('PC1', ascending=False).head(10))
print(loadings.sort_values('PC1').head(10))

loadings.plot(kind='bar', figsize=(12,4)); plt.show()
sns.heatmap(loadings, cmap='coolwarm', center=0); plt.show()
```

Table 6: Loadings of PC1, PC2 and PC3

Figure 9: Loadings of the first three components

12. Graphs and Results

The table below lists every graph and output to be included from the Colab notebook.

No.	Graph / Output	Purpose
1	Closing price trend	Shows how prices move over time
2	Daily return histogram	Shows spread and extreme returns
3	Correlation heatmap	Checks that stocks are related, so PCA is suitable
4	Scree plot	Helps choose k using the elbow
5	Cumulative variance plot	Shows variance kept by k components
6	Loadings heatmap / bar plot	Shows which stocks drive each component

Table 7: Graphs and their purpose

Key results (fill from Colab): number of stocks $p = \underline{\hspace{2cm}}$; components chosen $k = \underline{\hspace{2cm}}$; variance explained by PC1 = $\underline{\hspace{2cm}}\%$; by PC1 to PCk = $\underline{\hspace{2cm}}\%$.

13. Interpretation of Risk Factors

The table gives the usual meaning of the first components in equity data. Confirm each row against the actual loadings from the notebook before writing the final interpretation.

Component	Typical loading pattern	Risk factor meaning
PC1	Most stocks have the same sign and similar size	Market-wide (systematic) risk. All stocks rise and fall together.
PC2	Some groups of stocks positive, others negative	Sector or style contrast, for example one group of industries against another.

Component	Typical loading pattern	Risk factor meaning
PC3 and later	A small group of stocks stands out	Sub-sector or stock-specific risk; smaller variance.

Table 8: Typical interpretation of components

Practical use: the portfolio manager can watch a few factor scores instead of every stock. A portfolio with a large exposure to PC1 is exposed mainly to the market. A portfolio can be diversified by holding stocks with high loadings on different components. Components beyond k mostly represent noise and can be ignored.

Interpretation from our results (write from Colab):

_____.

14. Conclusion and Future Scope

Conclusion

PCA successfully reduced many correlated stock returns to a few uncorrelated principal components. The first component acts as the market-wide risk factor, and later components describe sector-level effects. Monitoring a small number of factors is simpler and easier to explain than tracking every stock. In this project, $k = \underline{\hspace{1cm}}$ components explain $\underline{\hspace{1cm}}\%$ of the total variance (fill from Colab).

Future scope

- Rolling-window PCA to study how risk factors change over time.
- Sector-wise PCA and use of a larger stock universe.
- Value-at-Risk (VaR) estimation using principal components.
- Sparse PCA and Kernel PCA for easier interpretation and non-linear structure.
- Autoencoders for non-linear dimensionality reduction.
- Portfolio optimisation based on factor exposures.

15. References

- [1] Jolliffe, I. T., and Cadima, J. (2016). Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A*, 374(2065), 20150202.
- [2] Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91.
- [3] Plerou, V., Gopikrishnan, P., Rosenow, B., Amaral, L. A. N., Guhr, T., and Stanley, H. E. (2002). Random matrix approach to cross correlations in financial data. *Physical Review E*, 65, 066126.
- [4] Jolliffe, I. T. (2002). *Principal Component Analysis* (2nd ed.). Springer.

Note: add the specific research paper suggested by the guide, if different.

Appendix: Viva Questions and Answers

Question	Answer
What is PCA?	A method that converts correlated variables into fewer uncorrelated components that keep most of the variance.
Why use returns and not prices?	Prices trend and have different scales. Returns are comparable across stocks.
Why standardize?	So that stocks with larger variance do not dominate the components.
What is an eigenvalue here?	The variance captured along the direction of its component.
What is a loading?	The weight of a stock in a principal component.
Why is PC1 usually the market factor?	Most stocks move together, so the largest shared variance is market-wide.
How did you choose k?	Using the scree plot, the eigenvalue > 1 rule and the cumulative variance threshold.
Why clip outliers instead of deleting?	Clipping keeps every trading day while limiting the pull of extreme values.
What are limitations of PCA?	It is linear, sensitive to outliers and scaling, and components can be hard to interpret.